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An Autonomous Institute



Department of Computer Science & Engineering

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Title of Project– An Explainable Hybrid Intelligence Framework for Static and Dynamic Phishing Website Analysis

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Introduction

Phishing websites have evolved into sophisticated cyberattacks that employ dynamic web content, JavaScript execution, and multi-stage redirections to evade conventional detection mechanisms. While machine learning has improved detection accuracy, most existing solutions rely primarily on static website features—such as URLs, HTML structure, domain information, and SSL metadata—and operate as uninterpretable black-box models. Consequently, these approaches fail to capture malicious runtime behaviors and provide insufficient evidence to justify their predictions, which limits analyst trust and practical adoption.

To address these gaps, this research introduces PhishScope, an explainable hybrid intelligence framework that combines multi-source static feature analysis with isolated browser-based dynamic execution. By integrating hybrid machine learning with Explainable Artificial Intelligence (XAI), PhishScope generates transparent, evidence-based risk assessments to improve detection reliability and support real-world cybersecurity deployment through practical investigation tools.

Research Objectives

- 01 Static Feature Extraction:** Extract and integrate multi-source static website features including URL structure, HTML content, Domain metadata, and SSL certificates.
- 02 Hybrid Intelligence Framework:** Design a hybrid intelligence framework combining comprehensive static analysis with dynamic browser-based behavioral monitoring.
- 03 Dynamic Behavioral Monitoring:** Monitor complex runtime behaviors (including JavaScript execution, live DOM modifications, and active network requests) within an isolated browser environment.
- 04 Explainable AI Integration:** Incorporate robust Explainable AI methods (SHAP / LIME) to deliver highly transparent, interpretable, and evidence-based risk predictions.
- 05 Framework Evaluation:** Evaluate the proposed framework's overall accuracy, latency, and practical deployability using standard real-world cybersecurity evaluation metrics.

Literature Review

Paper Title	Year	Feature(s) Used	Model/Technique	Identified Limitation
PhishParrot: A User Profile-Optimizing Crawler Leveraging LLMs	2026	URL, Dynamic Content	LLM-based Crawler	No XAI; limited static feature integration
PhishLumos: From a Single URL to Campaign-Level Mitigation	2026	URL, IP, Infrastructure	Campaign Analysis	No browser behavior analysis
Adaptive Cloud–Edge Coordination for Real-Time Phishing URL Detection	2026	URL	DistilBERT + MLP	Static URL-focused; no runtime behavior
XGBoost-Based URL Phishing Detection with Cross-Dataset Validation	2026	URL	XGBoost	URL-only analysis
DPMLF: URL Character and HTML Semantic Deep Features	2026	URL, HTML	CNN-based DL	No runtime behavior
Generalized Explainable AI Framework for Phishing Detection	2026	URL, Text	RF + CNN + SHAP	No browser execution analysis

Literature Review

Paper Title	Year	Feature(s) Used	Model/Technique	Identified Limitation
Whale Optimization and AutoML for Precise Phishing Detection	2025	URL	AutoML	No dynamic analysis
Convolutional Graph Network-Based Feature Extraction	2025	URL	GraphSAGE + SVM	URL-centric approach
Phishing URL Detection with Contextual Embeddings	2025	URL	Hybrid ML	Limited feature scope; no dynamic analysis
Impact of Feature Engineering in Phishing Detection	2025	URL, HTML	Random Forest	Lack of Explainable AI (XAI)
Explainable ML for Phishing Detection	2025	URL, Domain	XGBoost + SHAP	No HTML or dynamic behavioral features
Deep Learning-Based Website Detection	2024	HTML	CNN	Ignores SSL, Domain, and network execution

Challenges and Research Gaps

Core Challenges

1. Rapid Evolution of Phishing Techniques

Websites continually evolve URLs, content, and scripts, neutralizing static rules.

2. Limited Feature Coverage

Existing detectors neglect crucial HTML, SSL, network, and behavioral markers.

3. Dynamic Website Behavior

Redirects, JS actions, and network calls are completely missed by static-only tools.

4. Black-Box Machine Learning Models

Accurate classifiers offer little explanation or interpretable logic for predictions.

5. Dataset and Generalization Challenges

Models trained on specific environments fail when deployed to unseen campaigns.

6. Real-Time Detection Constraints

Evaluating execution behavior increases latency, creating deployment friction.

Identified Research Gaps

G1 – Lack of Unified Multi-Source Analysis

Studies focus on isolated features (URLs or HTML) rather than combining URL, HTML, domain, SSL, network, and behavior.

G2 – Limited Integration of Static and Dynamic Analysis

Predominantly static classifications fail to incorporate evidence from controlled dynamic website execution.

G3 – Insufficient Explainability

High-performing deep learning models lack explicit transparency, reducing analyst trust.

G4 – Limited Behavioral Evidence in Detection

Interactive elements and active behaviors are rarely used as classification features.

G5 – Generalization Across Datasets

Cross-dataset evaluation is limited, leaving true real-world robustness unproven.

G6 – Accuracy–Interpretability–Efficiency Trade-off

Prior research neglects balancing computational load and explainability with raw accuracy.

Research Gap Addressed by Our Work

Our research addresses these challenges and gaps by introducing a comprehensive framework that integrates: Multi-source Static Analysis, Hybrid ML, Isolated Dynamic Execution, Behavioral and Network Analysis, and Explainable AI.

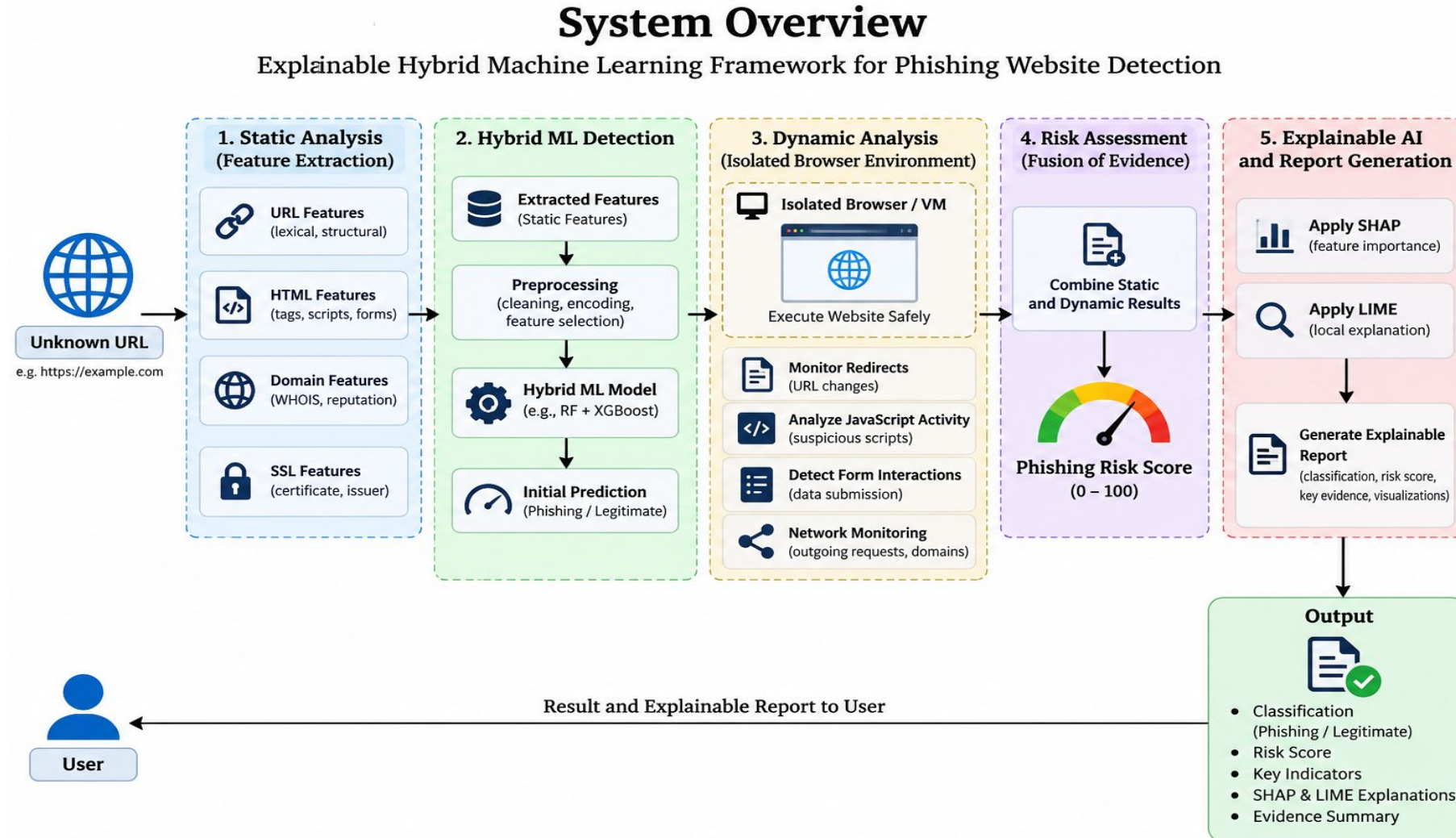
Problem Statement

Existing phishing website detection techniques often rely primarily on limited static features such as URLs, making them less effective against sophisticated phishing websites. They also provide limited insight into dynamic website behavior and often lack explainability. Therefore, there is a need for an integrated framework that combines **multi-source static features, dynamic behavioral analysis, hybrid machine learning, and Explainable AI** to achieve accurate and interpretable phishing website detection.

Proposed Solution

- **Multi-Source Feature Extraction** – Extract URL, HTML, domain, and SSL features.
- **Hybrid Machine Learning Detection** – Classify websites as phishing or legitimate using a hybrid ML model.
- **Isolated Dynamic Analysis** – Safely execute websites in an isolated browser environment and monitor their behavior.
- **Risk Score Generation** – Combine static, ML, behavioral, and network evidence to calculate a phishing risk score.
- **Explainable AI Report** – Use SHAP and LIME to explain the prediction and generate an interpretable detection report

System Overview



Methodology

- ❑ **Data Collection** – Collect phishing and legitimate website samples from reliable datasets and sources.
- ❑ **Static Feature Extraction** – Extract URL, HTML, domain, and SSL-related features from each website.
- ❑ **Data Preprocessing & Feature Engineering** – Clean, normalize, select, and transform the extracted features for machine learning.
- ❑ **Hybrid ML Model Development** – Train and combine suitable machine learning classifiers to detect phishing websites.
- ❑ **Dynamic Website Analysis** – Execute websites in an isolated browser environment and monitor redirects, JavaScript activity, forms, and network behavior.
- ❑ **Feature Fusion & Risk Assessment** – Combine static ML predictions with dynamic behavioral and network evidence to calculate a phishing risk score.
- ❑ **Explainable AI Analysis** – Apply SHAP and LIME to identify and explain the features influencing each prediction.
- ❑ **Performance Evaluation** – Evaluate the framework using Accuracy, Precision, Recall, F1-score, ROC-AUC, and detection time, with comparison against baseline models.

Tools and Technology

- ❑ **Programming Language:** Python
- ❑ **Machine Learning:** Scikit-learn, XGBoost, Random Forest
- ❑ **Explainable AI:** SHAP, LIME
- ❑ **Web Analysis:** BeautifulSoup, Requests
- ❑ **Browser Automation:** Selenium / Playwright
- ❑ **Isolated Environment:** Virtual Machine (VM)
- ❑ **Network Monitoring:** Wireshark / network traffic logging
- ❑ **Data Processing:** Pandas, NumPy
- ❑ **Visualization:** Matplotlib
- ❑ **Development Environment:** Jupyter Notebook / VS Code
- ❑ **Dataset Sources:** PhishTank, OpenPhish, and legitimate website datasets

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